# Prabhavit Gupta

Frontier Agents Engineer | Forward Deployed Engineering, Production Agent Runtimes & AI Quality

+39 351 610 7807 | prabhavitg@gmail.com | linkedin.com/in/prabhavit-gupta | github.com/prab-gupta
| Venice, Italy (EU Permanent Resident | Open to Relocation to SF / NY)

## Executive Summary

- **Frontier Agent Engineering & Forward Deployed Delivery:** Software engineer with 3 years commercial software experience embedding directly with engineering organizations to architect, integrate, and deploy production AI agent systems in complex enterprise environments.
- **Agent Runtimes, Tool Execution & Orchestration:** Deep practical expertise developing agent execution loops, Model Context Protocol (MCP) servers, sandboxed tool harnesses, and context pipelines that enable LLMs to reason reliably over structured enterprise data and live systems.
- **Production AI Quality & Evaluation Harnesses:** Proven track record operationalizing quantitative eval harnesses (Langfuse, DeepEval, LLM-as-a-judge), golden test datasets, and prompt regression suites to detect and prevent quality regressions across evolving agent workflows.

## Technical Skills & Core Competencies

- **Frontier AI Agents & Orchestration:** Autonomous Agent Loops, Tool Calling, Model Context Protocol (MCP), ReAct & Plan-and-Execute Architectures, Human-in-the-Loop Workflows, Multi-Agent Systems, Grounding & Guardrails.
- **Production AI Quality & Evals:** Quantitative Evaluation Frameworks (Langfuse, DeepEval), Offline Benchmarks, Regression Test Suites, LLM-as-a-Judge, Synthetic Data Generation, Distributed Tracing, Hallucination Mitigation.
- **Full-Stack & Backend Engineering:** Python (AsyncIO, FastAPI, PyTest), TypeScript, Next.js, Node.js, RESTful APIs, GraphQL, PostgreSQL, SQLite, Redis, Structured Outputs (Pydantic).
- **Enterprise Integration & DevOps:** Docker, Kubernetes, Linux CLI, GitOps / GitHub Actions CI/CD, Enterprise Auth/SSO, API Gateway Integrations, Technical Advisory & Field Enablement.

## Professional Experience

**Forward Deployed AI Engineer & Technical Solutions Consultant** • *Independent Consultancy*
2024 – Present • Venice, IT (Remote / Hybrid)

- Embedded directly with client engineering teams to scope technical requirements, design scalable production agent architectures, and deliver custom enterprise AI integrations.
- Built production agent runtimes and tool-calling harnesses using Model Context Protocol (MCP), enabling autonomous agents to safely inspect code repositories, query databases, and execute workflows.
- Implemented automated evaluation pipelines using Langfuse and custom regression suites, benchmarking tool parameter accuracy, agent reasoning paths, and latency under production traffic.
- Architected human-in-the-loop review checkpoints and automated fallback strategies for sensitive actions, meeting enterprise requirements for security, governance, and auditability.
- Abstracted recurring field deployment patterns into reusable client SDKs, integration templates, and architectural playbooks.

**Lead Developer & Product Engineer** • *FlowSpace.co*
Mar 2024 – Aug 2026 • London, UK (Remote)

- Led platform engineering and integration architecture, scaling the core platform to support 4,500+ active users across European tech markets.
- Re-architected core backend services from legacy infrastructure onto Next.js and API microservices, ensuring continuous operational uptime and strict data schema enforcement.
- Built automated data ingestion and extraction pipelines aggregating unstructured venue inventory across multiple European platforms with validation barriers.
- Integrated external third-party enterprise management APIs (OfficeRnD, Nexudus) for automated booking orchestration via real-time webhooks.
- Deployed automated conversational agent workflows connected to live operational databases, reducing manual triage workload while maintaining high resolution accuracy.

**Full-Stack Solutions Engineer** • *Tech Automation Systems*
2023 – 2024 • Milan / Remote, IT

- Built custom API integrations and automated backend microservices in Python and TypeScript, bridging client platforms with internal enterprise data warehouses.
- Implemented containerized automated testing suites within Docker CI/CD pipelines, establishing reliable integration baselines.

- Partnered directly with technical and business stakeholders to translate complex business processes into robust software specifications.

## Key Engineering Projects

- **Production AI Agent Evaluation & Tool Harness (TypeScript + Python + MCP + Langfuse):** Architected an execution harness enabling LLMs to safely invoke external tools over MCP, validating arguments against strict schemas with automated regression tracking and LLM-as-a-judge scoring.
- **Universal Multi-Source Data Ingestion Engine (Python + Playwright + REST APIs):** Designed an automated data harvesting and enrichment pipeline parsing 900+ structured and semi-structured external sources with schema validation and error isolation.

## Education, Languages & Legal Status

- **Bachelor of Arts in Philosophy, International Studies & Economics** – Ca' Foscari University of Venice (2020 – 2024)
  *Rigorous foundation in formal logic, epistemology, decision theory, and complex systems analysis.*
- **Languages:** English (C1 Full Professional / Fluent), Italian (Working proficiency), Hindi (Native).
- **Legal Status & Availability:** EU Long-Term Resident (Permesso di Soggiorno UE a Tempo Indeterminato, Italy). Open to international relocation to San Francisco or New York with visa sponsorship; available immediately.